Two Simulations for Mystery Tokens

AP Statistics · Unit 3 · Topic 3.14 · TEACHER KEY

CED TETHER

- **VAR-1.J** Identify questions suggested by variation between observed and expected counts in categorical data. **[Skill 1.A]**
- **VAR-1.J.1** Variation between what we find and what we expect to find may be random or not.

Essential question: What must a simulation vary to decide whether observed categorical discrepancies are surprising?

DOK-3 skill: 4.B · **FRQ pattern:** `fixed-count-vs-null-generation`

DOK rationale: Part (c) coordinates the null model, what each simulation preserves, the chi-square tail, and an evidence threshold to adjudicate two designs and explain the false assurance produced by the rejected design.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **varied-counts** (required) — Justifies Amara using new counts under the claimed probabilities
- **fixed-counts** (required) — Explains that Leo redraws the same counts and cannot model their variation
- **bounded-conclusion** (required) — Uses $0.186 > 0.05$ to call the result unsurprising without claiming proof

E	Explains both simulation mechanisms and uses the tail proportion for a bounded conclusion.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] THE PROBLEM — annotated

Suppose a company claims 10,000 mystery tokens are 50% red, 30% teal, and 20% gold. An SRS of 80 has the shown counts.

Leo: "Mix and redraw these same 80 tokens without replacement; recompute χ^2 for 1,000 trials."

Amara: "For each trial, generate 80 independent colors with probabilities 0.50, 0.30, and 0.20, then recompute χ^2." In Amara's 1,000 trials, 186 statistics are at least the observed value.

Token sample

Color	Claim	Observed
Red	0.50	32
Teal	0.30	30
Gold	0.20	18

FIRST TAKE — one sentence before you work the parts

Which plan can produce a different number of red tokens on the next trial? Explain in one sentence.
Key: Ungraded commitment; accept any evidence-based first impression.

[BOOK] EXPECTED COUNTS AND SIMULATION (keep on the page)

For a model probability p_i and sample size n, the expected count is $E_i = np_i$. The chi-square statistic adds the relative squared gaps: $\chi^2 = \sum(O-E)^2/E$. To model chance variation, generate new samples of the same size using the claimed probabilities. Compute the statistic each time. The fraction at least as large as the observed statistic estimates how often such a mismatch occurs if the model is true. A fraction at or below a stated cutoff is evidence against the model; a larger fraction does not prove the model.

DOK 1 **(a)** Find the three expected color counts from the company's percentages.

Key: $80(0.50) = 40$ red, $80(0.30) = 24$ teal, and $80(0.20) = 16$ gold.

DOK 2 **(b)** The observed statistic is $\chi^2 = 3.35$. Check the red contribution $(32-40)^2/40$, then compute the simulation proportion 186/1,000.

Key: The red contribution is $(32-40)^2/40 = 1.60$. The simulated upper-tail proportion is $186/1{,}000 = 0.186$.

DOK 3 **(c)** In 3–4 sentences, choose a simulation and justify what varies from trial to trial. Use the simulated proportion to decide whether the result is unusual at a 5 percent cutoff, and explain why this does not prove the company's claim.

Key: Amara is justified because each trial generates new counts using the claimed color probabilities. Leo redraws all of the same 80 tokens, so the counts and $\chi^2 = 3.35$ never change; his output cannot show sample-to-sample variation. Since $0.186 > 0.05$, the observed result is not unusual by the stated cutoff. This leaves chance variation plausible but does not prove the company's color claim.